

Containerization, Docker, and Docker Hub

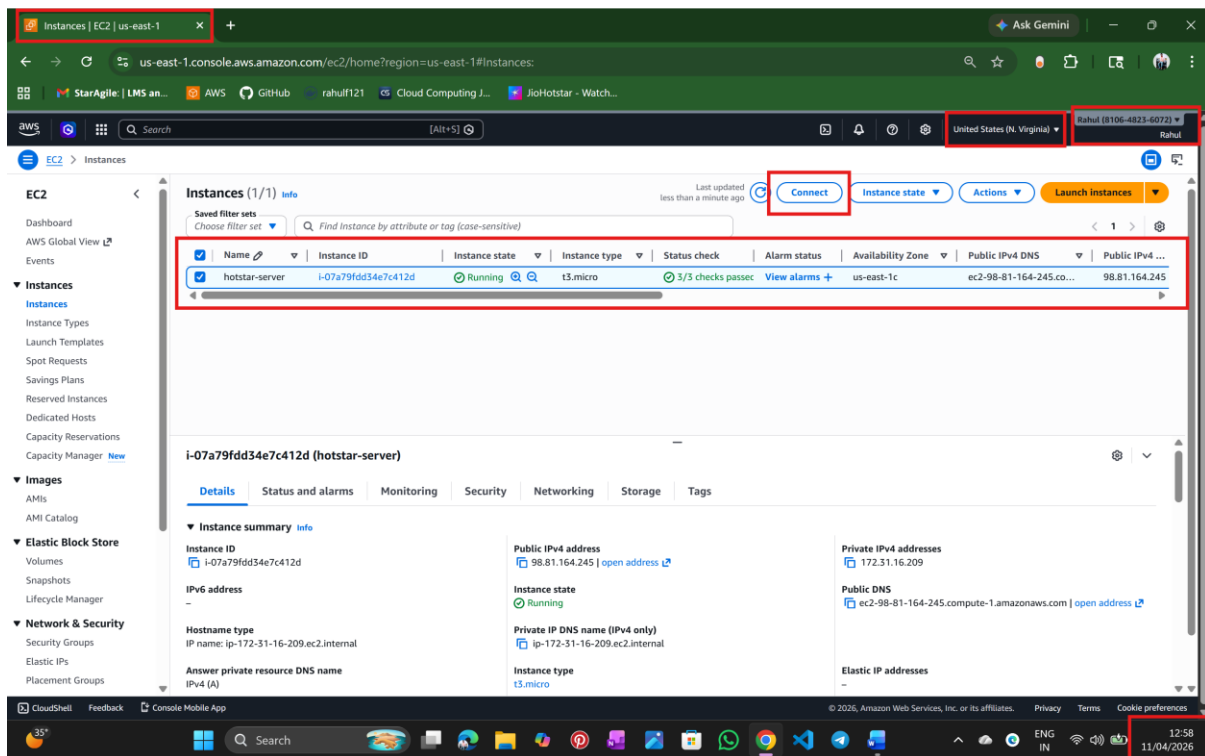
Submitted to - Vikul Sir

Date of Submission – 11/APR/2026

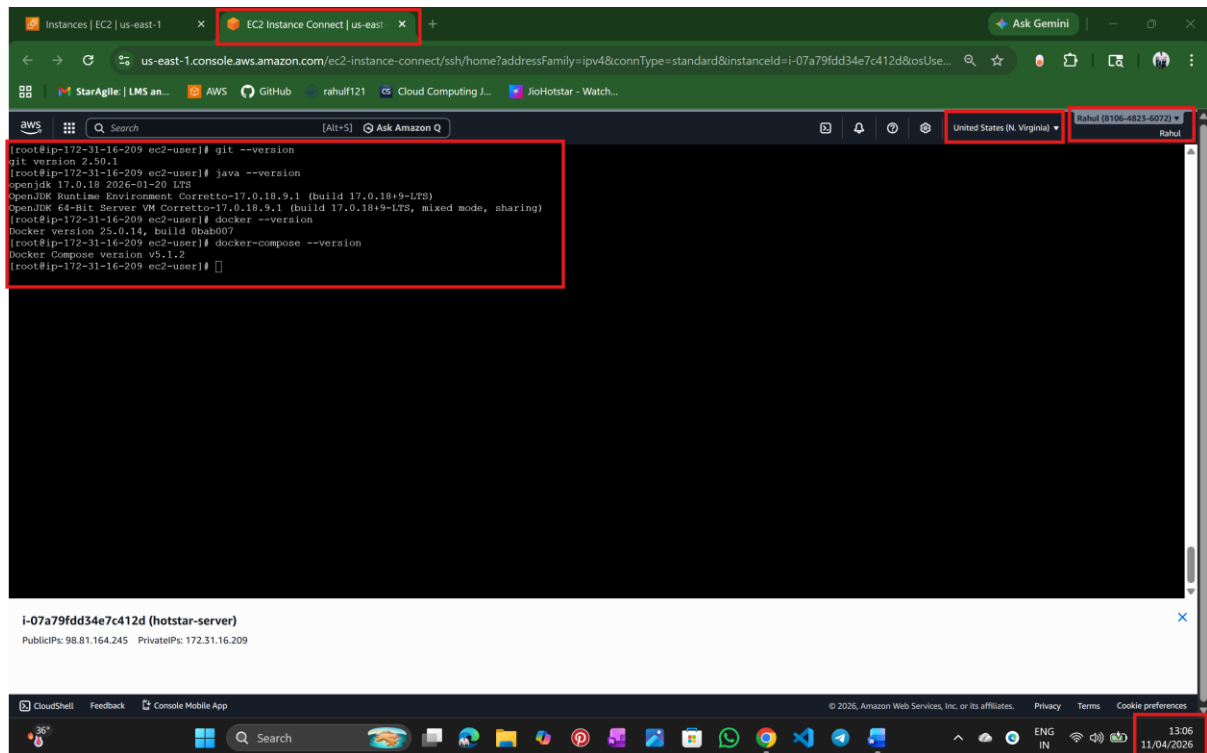
Submitted by – NALATHATI RAHUL

L3 - Demonstrate Docker Compose using the Application Image and MySQL Image to start and stop all container services

Step:1 login into AWS and launch an ec2 server



Step:2 install Prerequisites

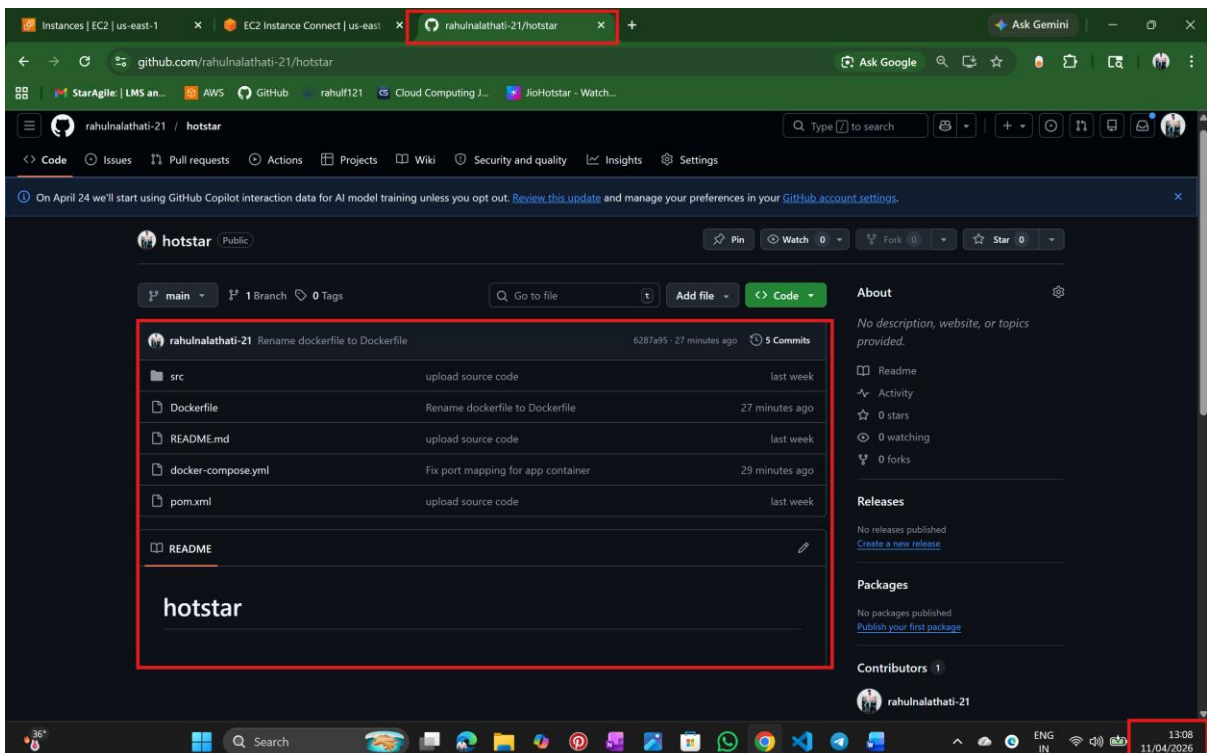


The screenshot shows the AWS Management Console with the 'EC2 Instance Connect' tab selected. The terminal output displays the following commands and their results:

```
[root@ip-172-31-16-209 ec2-user]# git --version
git version 2.50.1
[root@ip-172-31-16-209 ec2-user]# java --version
openjdk 17.0.18 2026-01-20 LTS
OpenJDK Runtime Environment Corretto-17.0.18.9.1 (build 17.0.18+9-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.18.9.1 (build 17.0.18+9-LTS, mixed mode, sharing)
[root@ip-172-31-16-209 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-16-209 ec2-user]# docker-compose --version
Docker Compose version v5.1.2
[root@ip-172-31-16-209 ec2-user]#
```

The instance details at the bottom show the instance ID 'i-07a79fdd34e7c412d (hotstar-server)' with public IP '98.81.164.245' and private IP '172.31.16.209'.

Step:3 Make sure Docker file and docker-compose file in git hub repo



The screenshot shows the GitHub repository page for 'hotstar' by user 'rahulnalathati-21'. The repository is public and has 5 commits. The file list includes:

- src
- Dockerfile
- README.md
- docker-compose.yml
- pom.xml

The 'About' section on the right indicates that no description, website, or topics are provided. The 'Releases' and 'Packages' sections also show no published content.



rahulnalathati-21 Rename dockerfile to Dockerfile

Code

Blame

9 lines (6 loc) · 191 Bytes

```
1 FROM tomcat:9.0
2
3 # Remove default apps (optional but clean)
4 RUN rm -rf /usr/local/tomcat/webapps/*
5
6 # Copy your WAR file
7 COPY target/myapp.war /usr/local/tomcat/webapps/ROOT.war
8
9 EXPOSE 8080
```



rahulnalathati-21 Fix port mapping for app container

Code

Blame

30 lines (27 loc) · 500 Bytes

```
1 version: '3.8'
2
3 services:
4   app:
5     build: .
6     container_name: app_container
7     ports:
8       - "8080:8080"
9     depends_on:
10       - db
11     environment:
12       DB_HOST: db
13       DB_USER: root
14       DB_PASSWORD: root
15       DB_NAME: testdb
16
17   db:
18     image: mysql:8.0
19     container_name: mysql_container
20     restart: always
21     environment:
22       MYSQL_ROOT_PASSWORD: root
23       MYSQL_DATABASE: testdb
24     ports:
25       - "3306:3306"
26     volumes:
27       - db_data:/var/lib/mysql
28
29 volumes:
30   db_data:
```

Step:4 clone the git repo and change to directory

The screenshot shows the AWS CloudShell interface. At the top, the browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'rahulnathathi-21/hotstar'. The address bar shows the URL 'us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-07a79fdd34e7c412d&osUse...'. The terminal window displays the following commands and output:

```
[root@ip-172-31-16-209 ec2-user]# git clone https://github.com/rahulnathathi-21/hotstar.git
Cloning into 'hotstar'...
remote: Enumerating objects: 114, done.
remote: Counting objects: 100% (114/114), done.
remote: Compressing objects: 100% (100/100), done.
remote: Total 114 (delta 4), reused 104 (delta 0), pack-reused 0 (from 0)
Resolving objects: 100% (114/114), 9.09 MiB | 59.30 MiB/s, done.
Resolving deltas: 100% (4/4), done.
[root@ip-172-31-16-209 ec2-user]# cd hotstar/
[root@ip-172-31-16-209 hotstar]# ls
Dockerfile  README.md  docker-compose.yml  pom.xml  src
[root@ip-172-31-16-209 hotstar]#
```

Below the terminal, the instance details for 'i-07a79fdd34e7c412d (hotstar-server)' are shown, including PublicIPs and PrivateIPs. The bottom of the screen shows the AWS CloudShell interface with a search bar and a status bar indicating the time as 13:11 on 11/04/2026.

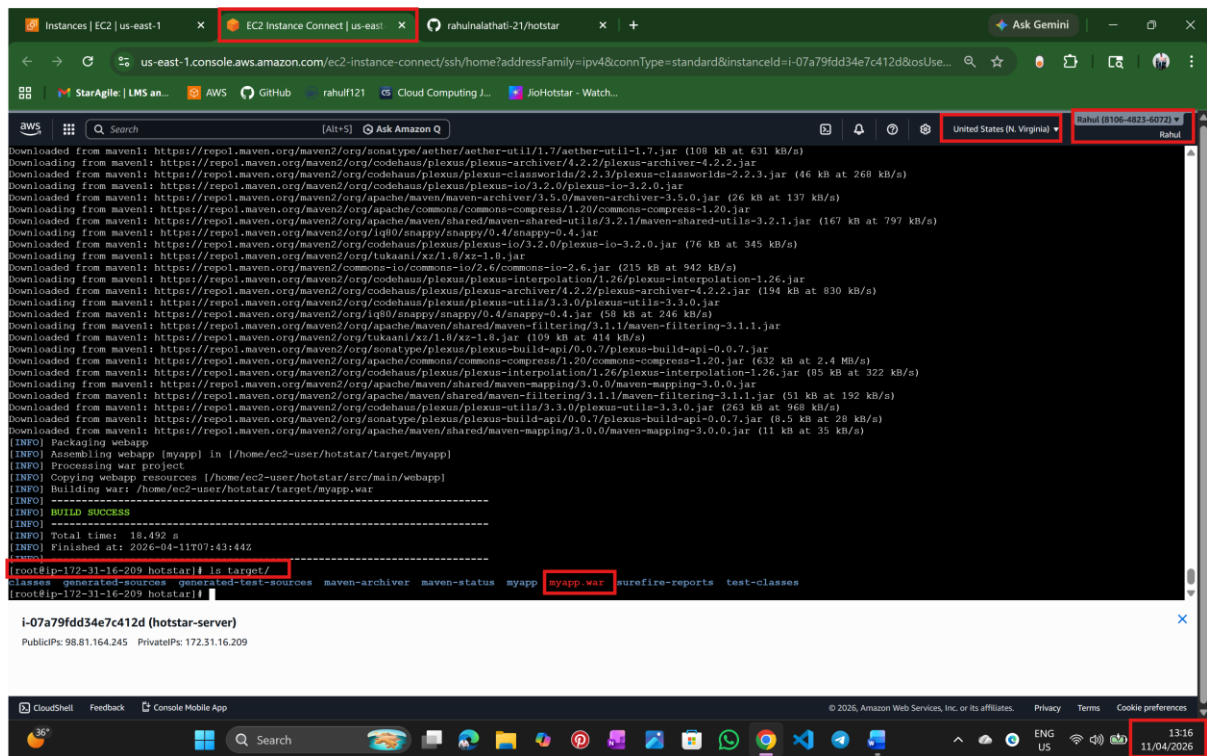
Step:5 Build WAR File

The screenshot shows the AWS CloudShell interface. At the top, the browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'rahulnathathi-21/hotstar'. The address bar shows the URL 'us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-07a79fdd34e7c412d&osUse...'. The terminal window displays the following commands and output:

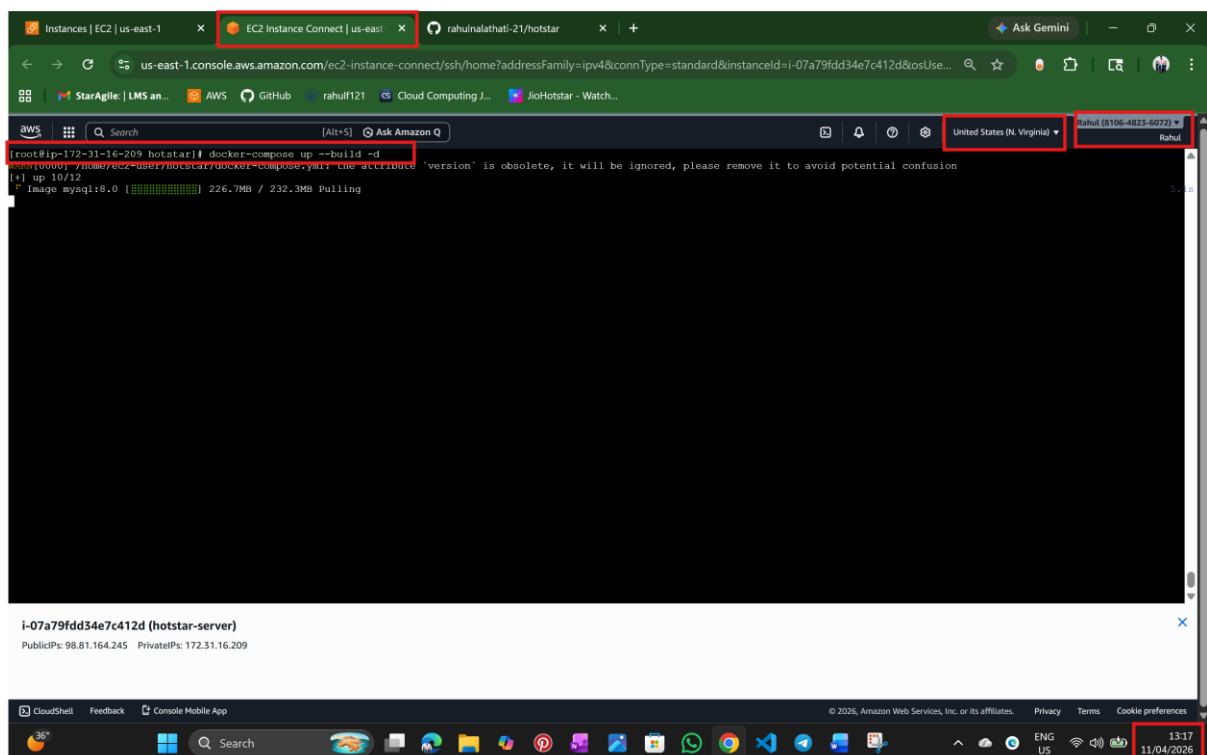
```
[root@ip-172-31-16-209 hotstar]# mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----K in.kastro:myapp-----
[INFO] Building myapp 8.3.3-SNAPSHOT
[INFO]
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clean-plugin-2.5.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clean-plugin-2.5.pom (3.9 kB at 8.5 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/22/maven-plugins-22.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/22/maven-plugins-22.pom (13 kB at 272 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/21/maven-parent-21.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/21/maven-parent-21.pom (26 kB at 447 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/apache/10/apache-10.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/apache/10/apache-10.pom (15 kB at 379 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clean-plugin-2.5.jar
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clean-plugin-2.5.jar (25 kB at 493 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-resources-plugin/2.6/maven-resources-plugin-2.6.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-resources-plugin/2.6/maven-resources-plugin-2.6.pom (8.1 kB at 225 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/23/maven-plugins-23.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/23/maven-plugins-23.pom (9.2 kB at 204 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/22/maven-parent-22.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/22/maven-parent-22.pom (30 kB at 633 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/22/maven-parent-22.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/22/maven-parent-22.pom (30 kB at 633 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-resources-plugin/2.6/maven-resources-plugin-2.6.jar
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-resources-plugin/2.6/maven-resources-plugin-2.6.jar (30 kB at 720 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.pom (10 kB at 255 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/24/maven-plugins-24.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-plugins/24/maven-plugins-24.pom (11 kB at 277 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/23/maven-parent-23.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/maven-parent/23/maven-parent-23.pom (33 kB at 693 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/apache/13/apache-13.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/apache/13/apache-13.pom (14 kB at 341 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.jar
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.1/maven-compiler-plugin-3.1.jar (43 kB at 895 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/2.12.4/maven-surefire-plugin-2.12.4.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/2.12.4/maven-surefire-plugin-2.12.4.pom (10 kB at 223 kB/s)
[INFO] Downloading from maven1: https://repo1.maven.org/maven2/org/apache/maven/surefire/surefire/2.12.4/surefire-2.12.4.pom
Downloaded from maven1: https://repo1.maven.org/maven2/org/apache/maven/surefire/surefire/2.12.4/surefire-2.12.4.pom (10 kB at 223 kB/s)
```

Below the terminal, the instance details for 'i-07a79fdd34e7c412d (hotstar-server)' are shown, including PublicIPs and PrivateIPs. The bottom of the screen shows the AWS CloudShell interface with a search bar and a status bar indicating the time as 13:14 on 11/04/2026.

Step:6 verify the target files



Step:7 Build & Start Containers



Step:8 check the containers list

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The main content area displays the 'Containers' tab for the EC2 instance 'i-07a79fdd34e7c412d (hotstar-server)'. A table lists the running containers:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
43ed25e06826	hotstar-app	"catalina.sh run"	30 seconds ago	Up 29 seconds	0.0.0.0:8080->8080/tcp, :::8080->8080/tcp	app_container
8f35f455bb6e	mysql:8.0	"docker-entrypoint.s..."	30 seconds ago	Up 29 seconds	0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 33060/tcp	mysql_container

Below the table, the instance details for 'i-07a79fdd34e7c412d (hotstar-server)' are shown, including Public IP: 98.81.164.245 and Private IP: 172.31.16.209. The bottom of the screen shows the Windows taskbar with the date and time '13:20 11/04/2026'.

Step:9 Access Your Application

The screenshot shows a web browser window displaying the Disney+ Hotstar website. The address bar shows the URL '98.81.164.245:8080'. The website header includes the Disney+ Hotstar logo, navigation links for TV, Movies, Cricket, Premium, Disney+, and Kids, and buttons for 'SUBSCRIBE' and 'LOGIN'. The main content area features a large banner for the Marvel Studios series 'Loki', with the text 'हिन्दी English' indicating language options. Below the banner, there are logos for Disney, Pixar, Marvel, Star Wars, and National Geographic. The bottom of the screen shows the Windows taskbar with the date and time '13:22 11/04/2026'.

Step:10 Stop All Services

The screenshot shows the AWS Management Console for an EC2 instance named 'hotstar-server' (i-07a79fdd34e7c412d) in the us-east-1 region. The terminal window displays the following commands and output:

```
[root@ip-172-31-16-209 hotstar]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
43ed25e06826   hotstar-app   "catalina.sh run"       30 seconds ago    Up 29 seconds    0.0.0.0:8080->8080/tcp, :::8080->8080/tcp    app_container
8f35f450bb6e   mysql:8.0     "docker-entrypoint.sh"   30 seconds ago    Up 29 seconds    0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 33060/tcp    mysql_container

[root@ip-172-31-16-209 hotstar]# docker-compose down
[WARN] [0000] /home/ec2-user/hotstar/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 3/3
  ✓ Container app_container   Removed
  ✓ Container mysql_container Removed
  ✓ Network hotstar_default   Removed

[root@ip-172-31-16-209 hotstar]# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED    STATUS    PORTS   NAMES
[root@ip-172-31-16-209 hotstar]# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED    STATUS    PORTS   NAMES
[root@ip-172-31-16-209 hotstar]#
```

The instance details show the public IP as 98.81.164.245 and private IP as 172.31.16.209.

The screenshot shows a web browser window with the address bar set to 98.81.164.245:8080. The page displays a message: "This site can't be reached" and "98.81.164.245 refused to connect." Below the message, it says "Try:" and lists two options: "Checking the connection" and "Checking the proxy and the firewall". The error code "ERR_CONNECTION_REFUSED" is shown at the bottom. A "Reload" button is visible.

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NALATHATI RAHUL